

Nguyen Khanh Duy — Software Engineer

Binh Trung, HCM • kduyyy193.for.dev@gmail.com • (+84) 985 308 170

WORK EXPERIENCE

Levinci Co., Ltd

Ho Chi Minh, Vietnam

Software Engineer | 01/2023 – Present

Internal AI Agent Platforms

- Designed **hybrid retrieval architecture** combining Knowledge Graph and vector search, optimized with caching and parallel processing to achieve ~1.5–2s time-to-first-token (TTFT) while maintaining ~85–90% retrieval accuracy.
- Built internal **multi-agent AI framework** with role-based workflows, reducing average task completion time by ~40% and achieving ~90% task success rate, enabling internal teams to collaboratively build projects with AI agents.
- Designed **multi-tenant architecture** and enterprise-grade license/role management, managing 5+ isolated workspaces with zero cross-tenant data incidents.
- Built **AI Assistant** for real-time chat support and fast information retrieval, applying the **AIDA framework** to guide user conversations toward action (guided conversion), with **interactive map rendering** to visualize location-based results, achieving ~2.5–3s average response time and ~85% query resolution accuracy.

POS Systems (Retail & F&B)

- Implemented **local state persistence** for cart/order data, allowing POS to continue operating during network interruptions with zero transaction loss, auto-syncing within seconds of reconnect.
- Built **optimistic UI updates** for cart and checkout actions, reducing perceived action latency from ~600ms to under 100ms while confirming with server in the background.
- Structured **frontend state management** to handle 30+ concurrent orders/tables without UI lag or state conflicts.

Social Platforms

- Built **timestamp-based product tagging** applied across 100+ masterclass videos, enabling in-video shopping tied to lesson context.
- Developed **CV builder with dynamic Canvas rendering**, generating resumes from user data across 5+ customizable templates.
- Optimized **news feed delivery** with Redis caching and cursor-based pagination, reducing database load by ~35% and cutting average feed load time from ~1s to ~400ms under high-traffic conditions.

ERP Systems

- (Construction) Built **cost-tracking** module comparing budgeted vs. actual spend per work item, surfacing overruns 2–3 weeks earlier, applied across 10+ construction projects.
- (Legal) Automated legal **deadline reminders**, reducing missed filing/procedural deadlines by ~80%.
- (Education) Built **room allocation system** for dormitory check-in/check-out cycles, automating bed assignment for 300+ dormitory rooms and cutting manual processing time by ~60%.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python.

Frameworks: NodeJS, NestJS, ReactJS, NextJS, FastAPI.

Databases: SQL, NoSQL, In-Memory, Vector, Graph DB.

AI/LLM: RAG (Retrieval-Augmented Generation), Knowledge Graph, Vector Search, Hybrid Retrieval Architecture, Multi-Agent Orchestration.

Tools: Docker, RabbitMQ, Git, Jenkins, Unit Testing.

EDUCATION

